Octal to Binary & Binary to Octal Conversion

>> There are eight different digits in octal number system.
>> Each one of them can be represented by a equivalent 3-bit number.

$r = 8$

0 to $(8-1)$

0 to 7

0 –	000
1 –	001
2 –	010
3 –	011
4 –	100
5 –	101
6 –	110
7 –	111

Octal to Binary:

i) $(37.45)_8 \longrightarrow (011111.100101)_2$

011 111 . 100 101

ii) $(22.07)_8 \longrightarrow (010010.000111)_2$

T — 1 1 1

Binary to Octal:

H.W. i) $(672.13)_8 \rightarrow (\quad)_2$

ii) $(10111011.1001)_2 \rightarrow (\quad)_8$

i) $(\underbrace{010}_{2}\underbrace{110}_{1}.\underbrace{110})_2 \longrightarrow (26.6)_8$

ii) $(\underbrace{000}\underbrace{001}\underbrace{001}.\underbrace{100})_2 \longrightarrow (011.4)_8$